



BITS Pilani
Pilani Campus

Blockchain Technology and Systems (SEZG569/SSZG569)

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Why a course on Blockchain?

What is all the excitement about?



- 1) **Basic application:** a digital currency (stored value)
 - Current largest: Bitcoin (2009), Ethereum (2015)
 - Global: accessible to anyone with an Internet connection

- 2) **Beyond stored value:** decentralized applications (DAPPs)
 - **DeFi:** financial instruments managed by public programs (examples: stablecoins, lending, exchanges,)
 - **Asset management (NFTs):** art, game assets, domain names.
 - **Decentralized organizations (DAOs):** (decentralized governance) DAOs for investment, for donations, for collecting art, etc.

- 3) **New programming model:** writing decentralized programs

What is a blockchain?

Abstract answer: a blockchain provides

- coordination between many parties,
- when there is no single trusted party

if trusted party exists $\Rightarrow$ no need for a blockchain

[financial systems: often no trusted party]

What is Blockchain?



☐ A Linked List

- ☐ Replicated
- ☐ Distributed
- ☐ Consistency maintained by Consensus
- ☐ Cryptographically linked
- ☐ Cryptographically assured integrity of data

☐ Used as

- ☐ Immutable Ledger of events, transactions, time stamped data
- ☐ Tamper resistant log
- ☐ Platform to Create and Transact in Cryptocurrency
- ☐ log of events/transactions unrelated to currency

What is a blockchain?



LAYER 3

User Interface (e.g., web3)

LAYER 2

Applications (Solidity, Move, Motoko)

LAYER 1.5

Compute Layer (blockchain computer)

LAYER 1

Consensus Layer

Layer 1: Consensus Layer (Informal)

A public append-only data structure:

achieved by replication

- **Persistence:** once added, data can never be removed*
- **Safety:** all honest participants have the same data**
- **Liveness:** honest participants can add new transactions
- **Open(?):** anyone can add data (no authentication)

LAYER 1

Consensus Layer

This Not a New Problem ...



State machine replication: studied since the 1980s

Google, Amazon, Banks, all have lots of servers:

- need to ensure state is consistent across all servers
- Known # of servers, and all are authorized.

New aspect: open consensus

- Anyone can write new data to the blockchain
- Was shown to be impossible! [Barak, Canetti et. al 05]
- Nakamoto [2008]
 - A way to bypass the bound using proof of work

Open Consensus: How?



PROOF-OF-WORK

First party to solve puzzle creates next block

- sybil resistant selection of a random party

Problems:

- slow, wastes energy

 **bitcoin**

 **ethereum**

PROOF-OF-STAKE

Fast block creation

No energy waste

But more complex

 **ethereum**

 **Tendermint**

 **DEFINITY**  **celo**

 **Algorand™**

PROOF-OF-SPACE

 **chia**

 **Filecoin**

MANY MORE IDEAS




SOLANA

Layer 1.5: The blockchain computer

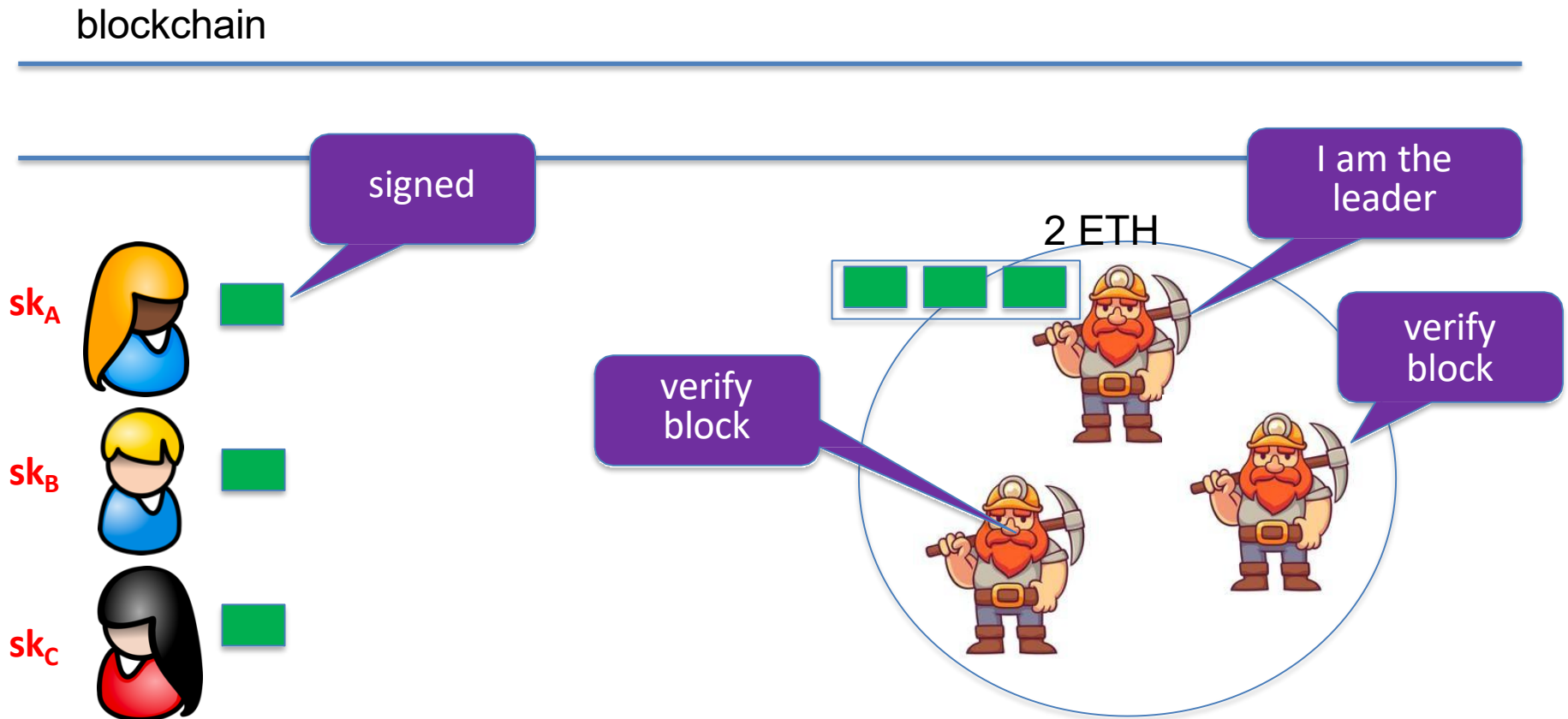


APP logic is encoded in a program that runs on blockchain

- Rules are enforced by a public program (public source code)
 - ⇒ **transparency**: no single trusted 3rd party
- The DAPP program is executed by parties who create new blocks
 - ⇒ **public verifiability**: everyone can verify state transitions

LAYER 1.5	Compute Layer (blockchain computer)
LAYER 1	Consensus Layer

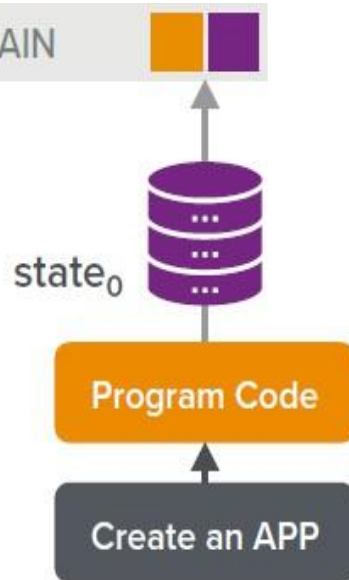
How are blocks added to chain?



Running Programs on a Blockchain (APPs)



BLOCKCHAIN



LAYER 1.5

Compute Layer (blockchain computer)

LAYER 1

Consensus Layer

Execution Environment



BITCOIN SCRIPT

LIMITED COMPUTING ENVIRONMENT

- Limited instruction set (no loops)
- Sufficient for some tasks:
 - atomic swaps
 - payment channels, ...

ETHEREUM

GENERAL PROGRAMMING ENVIRONMENT (SOLIDITY)

- EVM is a general purpose computing environment
- APP code updates internal state in response to transactions
- Calling APP costs fees (gas)
 - prevents DoS on miners
 - **storing on-chain state costs fees**

Decentralized Applications (DAPPs)

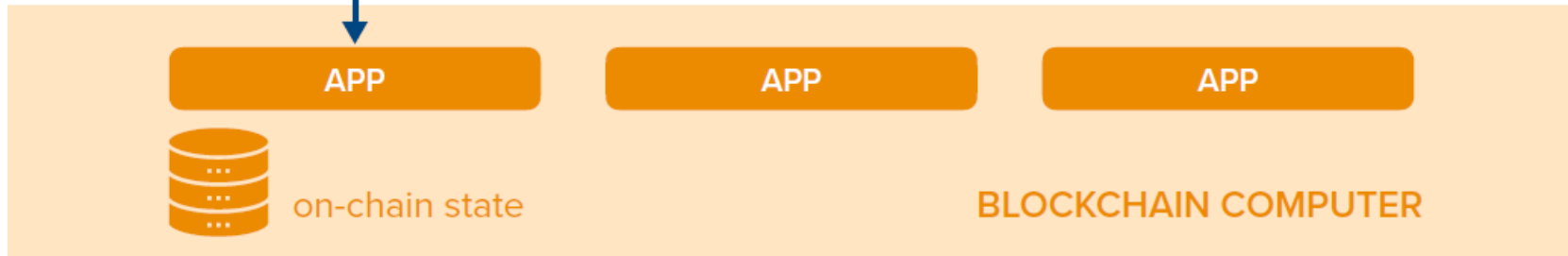
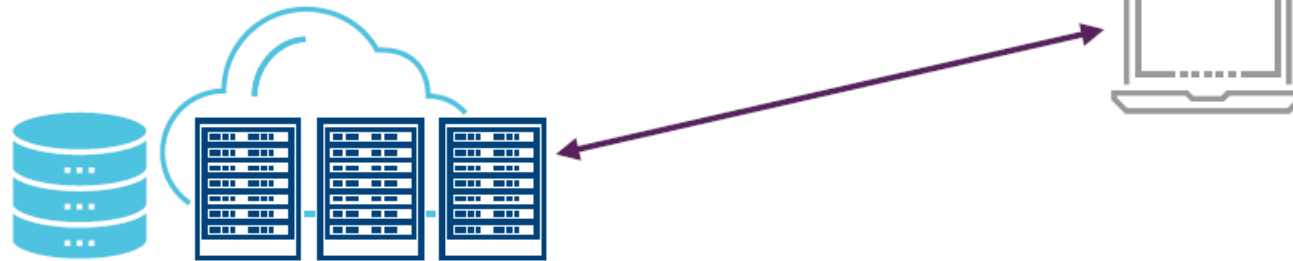


LAYER 2	Applications (Solidity, Move, Motoko)
LAYER 1.5	Compute Layer (blockchain computer)
LAYER 1	Consensus Layer

Common App Architecture



Layer 4: user facing servers



LAYER 1

Consensus Layer

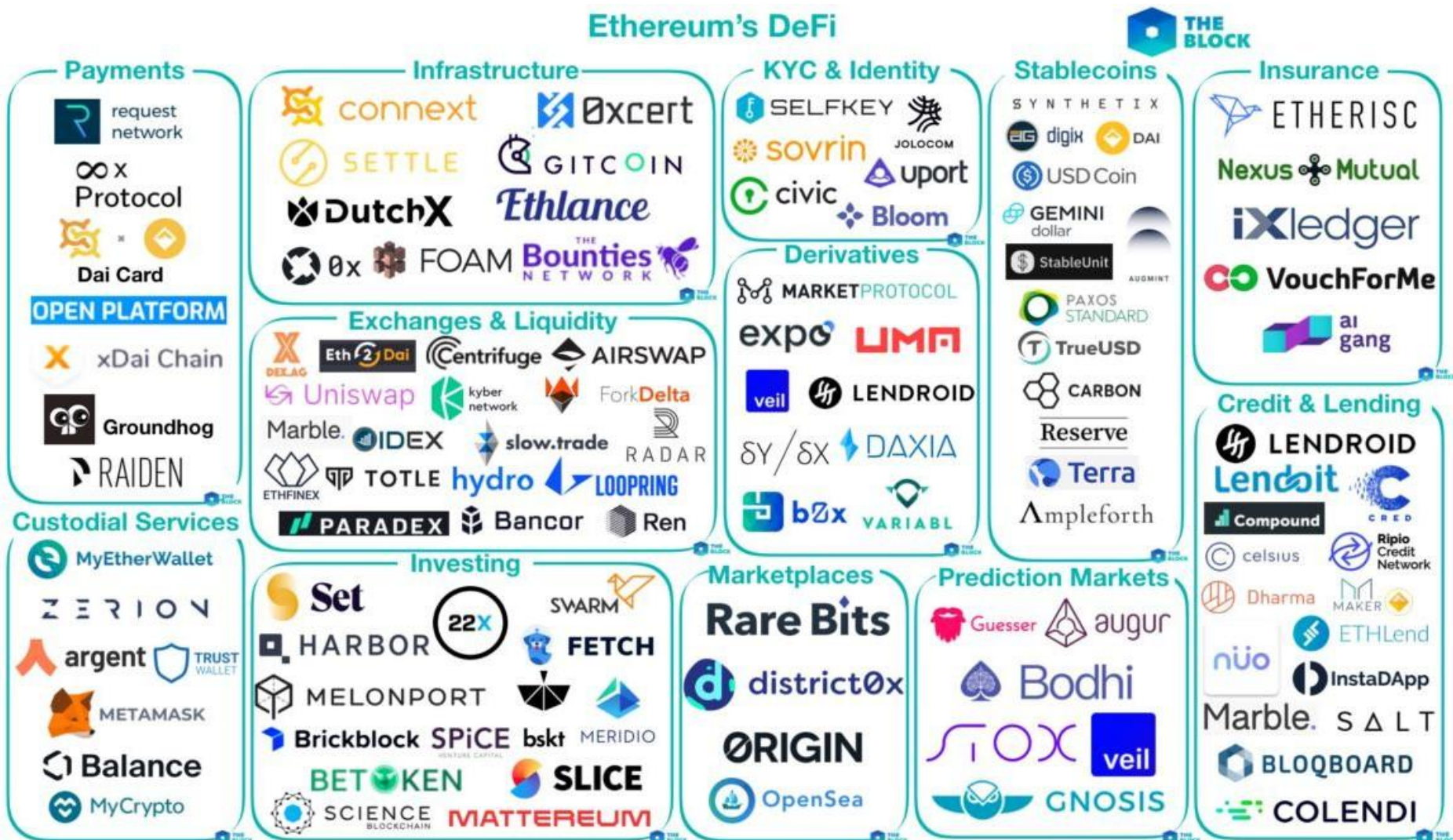
Ethereum DeFi

innovate

achieve

lead

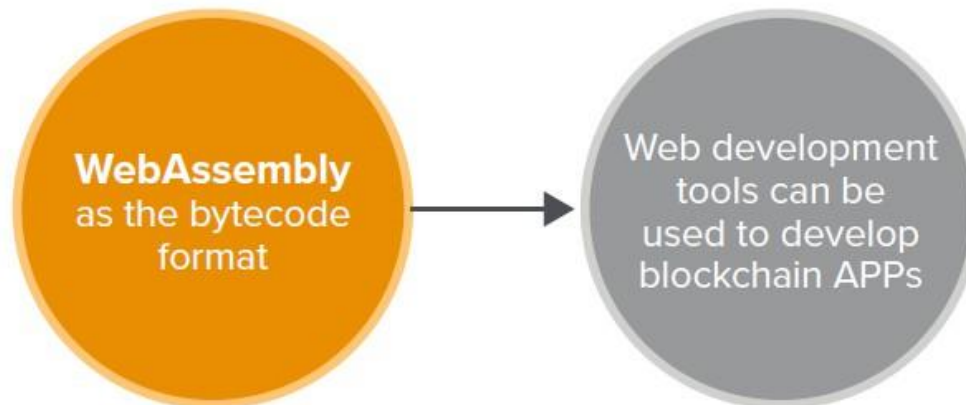
Ethereum's DeFi



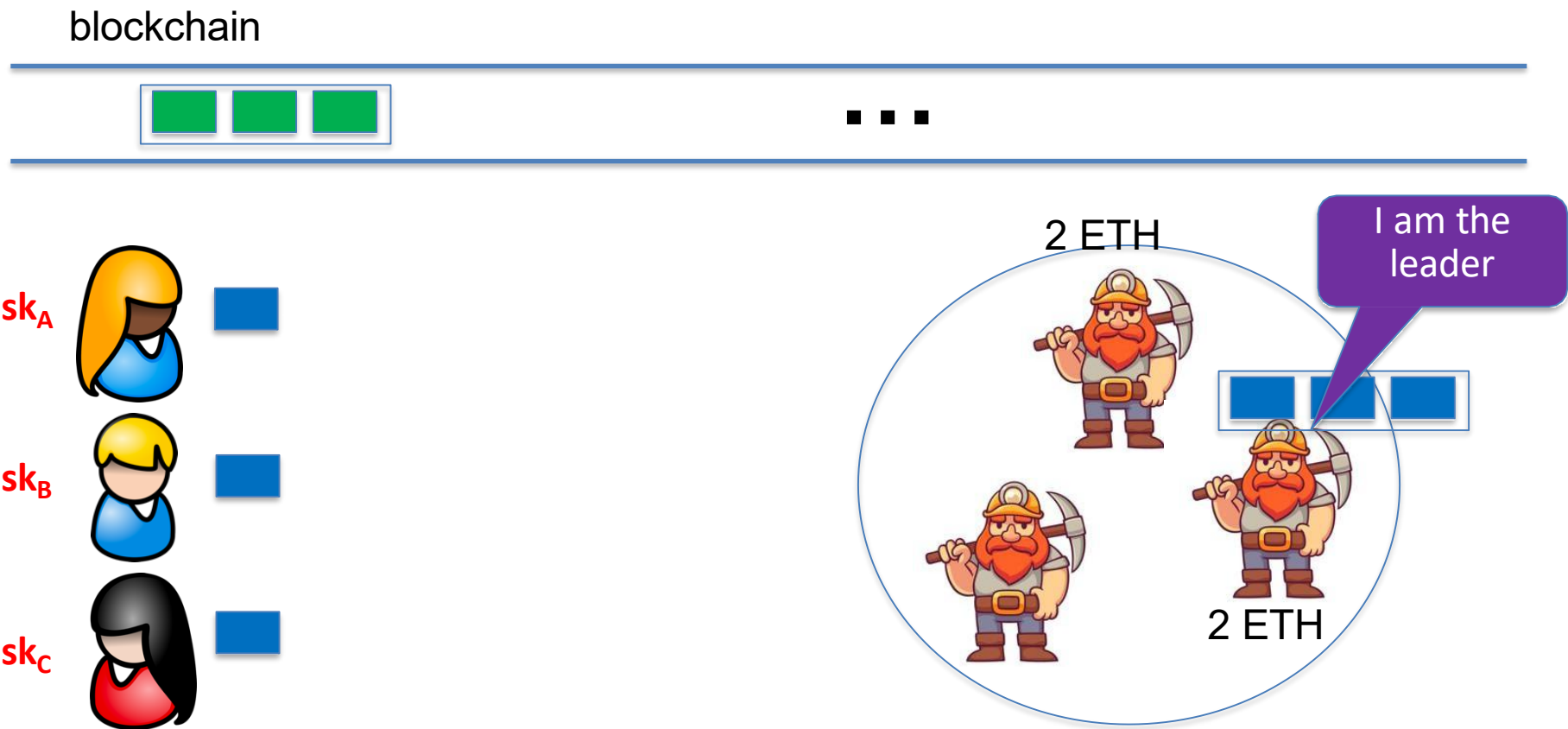
General Execution Environments



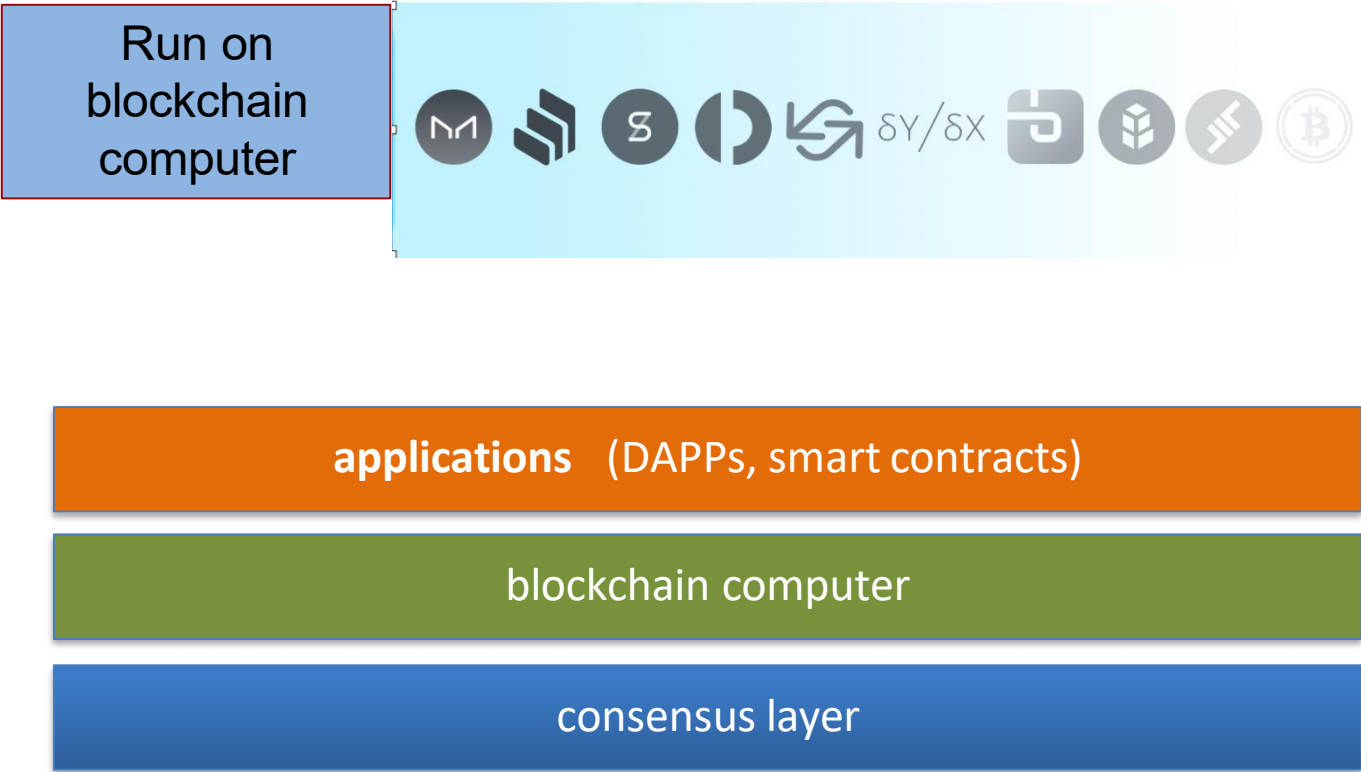
Recent projects



How are blocks added to chain?



Decentralized applications (DAPPS)



Common DAPP architecture



Top layer: user facing servers



end user

